

NETWORK SECURITY

MAC Addresses, ARP & IPv6

Deep Dive into Network Layer Addressing & Protocols

IEEE 802.3

ARP

IPv6

Layer 2

Cybersecurity

MAC

Media Access
Control

DE:AD:BE:EF:13:37

OUI (3 bytes)

NIC (3 bytes)

IEEE STANDARDS

802.3

Ethernet

802.15

Bluetooth

802.11

WLAN

WHAT IS A MAC ADDRESS?

48-bit

Physical address (6 octets in hex)

Unique

Set by manufacturer, changeable in software

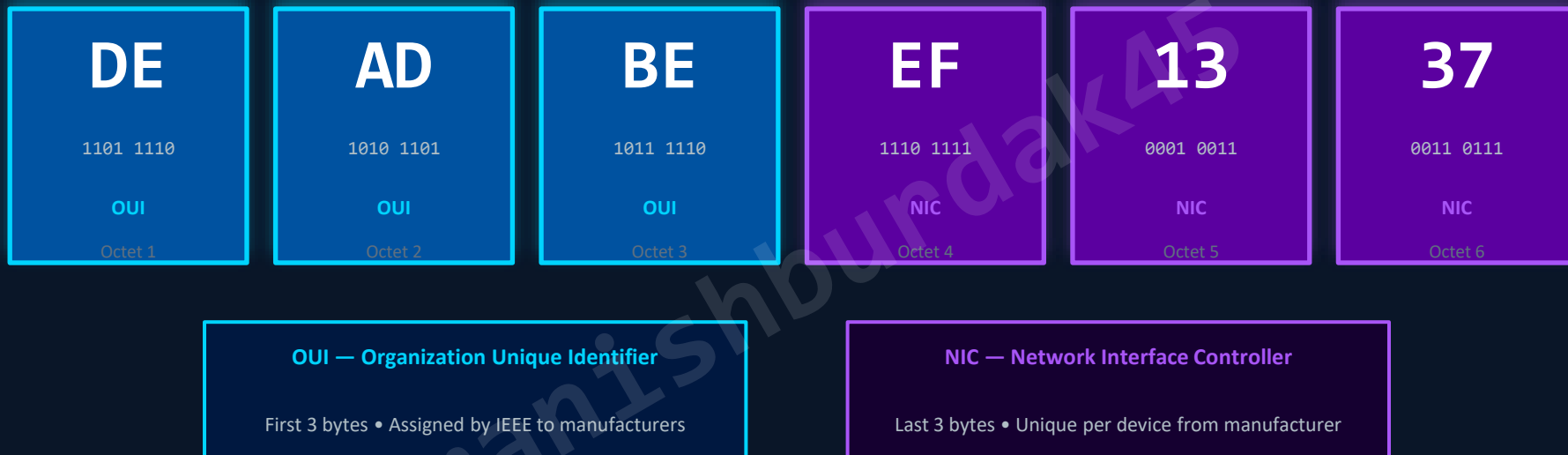
Layer 2

Operates at Data Link Layer (OSI)

Per NIC

Each network interface has its own MAC

MAC ADDRESS STRUCTURE



Notation Formats:

DE:AD:BE:EF:13:37

DE-AD-BE-EF-13-37

DEAD.BEEF.1337

MAC ADDRESS TYPES

1

UNICAST

One-to-One

Packet reaches exactly one specific device

Last bit of 1st octet = 0

DE:AD:BE:EF:13:37

2

MULTICAST

One-to-Many

Packet sent once; multiple devices can accept it

Last bit of 1st octet = 1

01:00:5E:EF:13:37

3

BROADCAST

One-to-All

Delivered to all devices on the LAN segment

All bits = 1

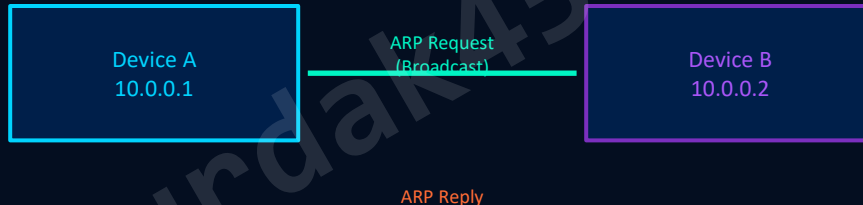
FF:FF:FF:FF:FF:FF

ADDRESS RESOLUTION PROTOCOL (ARP)

WHAT IS ARP?

Resolves Layer 3 IP addresses to Layer 2 MAC addresses.
Essential for communication within a Local Area Network.

ARP RESOLUTION FLOW



TSHARK OUTPUT

```
10.129.12.100 → 10.129.12.255 ARP Who has 10.129.12.101? Tell 10.129.12.100
10.129.12.101 → 10.129.12.100 ARP 10.129.12.101 is at AA:AA:AA:AA:AA:AA
```

ARP REQUEST

Broadcast to all devices — "Who has IP X?"

ARP REPLY

Unicast back — "IP X is at MAC AA:BB:CC..."

The diagram shows a network topology with three nodes: VICTIM (10.0.0.2), GATEWAY (10.0.0.1), and ATTACKER (CC:CC:CC:CC). A red line represents the network path. A blue line from the Victim to the Gateway is labeled 'ARP request'. A red line from the Attacker to the Victim is labeled 'Fake ARP Reply "GW is at CC:CC:CC"'. A red line from the Victim to the Attacker is labeled 'Traffic intercepted!'. A large diagonal watermark 'dak45' is visible across the diagram.

```
Attacker → Victim ARP 10.0.0.255 is at CC:CC:CC:CC:CC:CC (Fake gateway!)
Victim → Broadcast ARP Who has 10.0.0.1? Tell 10.0.0.2 (Querying poisoned gateway)
```

Alter MAC to match another device's identity

Intercept & relay traffic between victim & gateway

Overflow switch MAC table with fake addresses

Corrupt the victim's ARP table entries

MAC ATTACK VECTORS & DEFENSES

⚔️ ATTACK VECTORS

MAC Spoofing

Clone another device's MAC to bypass access control

MAC Flooding

Overwhelm switch CAM table → switch acts as hub

MAC Filtering Bypass

Spoof whitelisted MAC to gain network entry

ARP Cache Poisoning

Associate attacker MAC with legitimate IP

🛡️ DEFENSES

Network Segmentation

VLANs limit broadcast domains and lateral movement

Dynamic ARP Inspection

Validate ARP packets against DHCP snooping table

Strong Authentication

802.1X port-based NAC with certificates

IPSec / SSL

Encrypt traffic to thwart MITM interception

IPv4

Legacy

IPv6

Next Generation

Bit Length 128 bits

Address Space ~340 Undecillion

Representation Hexadecimal

Dynamic Address SLAAC / DHCPv6

IPsec Mandatory

NAT Required No (End-to-End)

Broadcast Replaced by Multicast

IPv6 ADDRESS FORMAT

FULL NOTATION:

fe80:0000:0000:0000:dd80:b1a9:6687:2d3b/64

SHORT NOTATION (RFC 5952):

fe80::dd80:b1a9:6687:2d3b/64

NETWORK PREFIX

fe80:: (first 64 bits)
Identifies the network/subnet/range
Default prefix length: /64

INTERFACE IDENTIFIER

dd80:b1a9:6687:2d3b (last 64 bits)
Derived from 48-bit MAC address
Converted to 64-bit EUI-64 format

RFC 5952 Rules:

All lowercase • Omit leading zeros • Shorten consecutive zero blocks with :: • Use :: only once

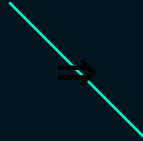
IPv6 ADDRESS TYPES



UNICAST

Single interface address. One sender to one specific receiver.

- Global Unicast (2000::/3)
- Link-Local (fe80::/10)
- Loopback (::1)



ANYCAST

Assigned to multiple interfaces. Delivered to the nearest (topology) single one.

- Used for load balancing
- Closest node responds
- No broadcast equivalent



MULTICAST

One packet to multiple interfaces. Replaces IPv4 broadcast entirely.

- All nodes (ff02::1)
- All routers (ff02::2)
- Solicited-node multicast

⚡ Note: IPv6 eliminates Broadcast — replaced entirely by Multicast addresses

IPv6 ADVANTAGES

Why the internet needs to move forward

2¹²⁸

340 Undecillion Addresses

Virtually unlimited — every device on Earth can have millions of IPs

Auto

SLAAC Auto-Configuration

Devices configure their own IP without DHCP via Stateless Address Autoconfiguration



Built-in IPsec Encryption

End-to-end encryption is mandatory, not optional like IPv4

Direct

No NAT Required

True end-to-end connectivity; each device gets a globally unique IP



Faster Routing

Simplified fixed header (40 bytes) vs IPv4 variable-length header

4 GB

Jumbo Frames

Data packets up to 4 GByte for large data transfers

HEXADECIMAL SYSTEM

The language of MAC addresses and IPv6

DECIMAL → HEX → BINARY					
0	0	0000	1	1	0001
2	2	0010	3	3	0011
4	4	0100	5	5	0101
6	6	0110	7	7	0111
8	8	1000	9	9	1001
10	A	1010	11	B	1011
12	C	1100	13	D	1101
14	E	1110	15	F	1111

IPv4 → HEX CONVERSION

192.168.12.160

Binary:

1100 0000

1010 1000

0000 1100

1010 0000

Hex:

C0

A8

0C

A0

Decimal:

192

168

12

160

Hex Result: C0.A8.0C.A0

OSI LAYER MAPPING

L7	Application	HTTP / DNS / SMTP	
L4	Transport	TCP / UDP	
L3	Network	IPv4 / IPv6 / ARP	★
L2	Data Link	MAC / Ethernet 802.3	★
L1	Physical	Cables / Wi-Fi / Bluetooth	

KEY TAKEAWAYS

- 01 MAC addresses are 48-bit physical addresses divided into OUI (manufacturer) and NIC (device) parts
- 02 ARP maps Layer 3 IP addresses to Layer 2 MAC addresses and is vulnerable to spoofing/poisoning attacks
- 03 IPv6 uses 128-bit hexadecimal addresses to solve IPv4 exhaustion — offering 340 undecillion unique IPs
- 04 MAC addresses can be spoofed — rely on strong authentication & network segmentation for security
- 05 IPv6 mandates IPsec, eliminates NAT, and replaces broadcast with multicast for modern networking